

APIs, Tooling & Deployment

ResNet Academy | Frontend Development

HTTP quick facts

Status	Meaning
200 201 204	OK, Created, No Content
301 304	Moved permanently, Not modified
400 401 403 404 422	Bad request, Not authenticated, Not allowed, Not found, Validation failed
500 503	Server error, Service unavailable

CORS errors are fixed on the **server** with **Access-Control-Allow-Origin**. An origin is scheme + host + port.

fetch patterns

```

async function getJson(url) {
  const response = await fetch(url);
  if (!response.ok) throw new Error(`HTTP ${response.status}`); // fetch does NOT reject on 404/500
  return response.json();
}

await fetch(url, {
  method: "POST",
  headers: { "Content-Type": "application/json" },
  body: JSON.stringify(data),
});

new URLSearchParams({ q: "css", page: 2 }).toString(); // "q=css&page=2"
const ctrl = new AbortController(); fetch(url, { signal: ctrl.signal }); ctrl.abort();
await Promise.all([getJson(a), getJson(b)]); // run in parallel

```

The four UI states

State	Show
Loading	An indicator. Disable repeat actions
Success	The data
Empty	A friendly "nothing found" message
Error	A plain message and a retry option. No stack traces

localStorage

```

localStorage.setItem("key", JSON.stringify(value));
const value = JSON.parse(localStorage.getItem("key")) ?? fallback;
localStorage.removeItem("key"); localStorage.clear();

```

- Strings only, roughly 5 MB per origin, synchronous, shared by tabs of one origin.
- **sessionStorage** is the same API but ends when the tab closes.
- Never store passwords, access tokens or personal data there.

ES modules

```

export function getJson(url) { ... } export default App;
import { getJson } from "./api.js"; import App from "./App.js";
<script type="module" src="main.js"></script> // deferred automatically

```

Local imports need **./** and the file extension. **file://** does not work, so use a dev server.

npm and Vite

```

npm create vite@latest my-app -- --template vanilla
npm install # install dependencies into node_modules
npm run dev # dev server, hot reload
npm run build # production build into dist/
npm run preview # test the production build locally
import.meta.env.VITE_API_URL // only VITE_-prefixed variables reach the browser

```

Git essentials

```
git init | git status | git add . | git commit -m "message"
git log --oneline | git diff | git switch -c feature-name
git remote add origin <url> | git push -u origin main | git pull
.gitignore: node_modules/  dist/  .env
```

Deployment checklist

Host	Settings
Netlify	Build command <code>npm run build</code> . Publish directory <code>dist</code> .
GitHub Pages	Set <code>base: "/repo-name/"</code> in <code>vite.config.js</code> . Publish <code>dist</code> .

- Test the live URL on a real phone in a private window.
- No failed requests in the Network panel. No secrets committed.
- Run Lighthouse. Write a README with the live link.